

Boao (James) Chen

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Research Interests

Databases, Data Systems, Storage Engines, Machine Learning for Systems

Education

Doctor of Philosophy, Computer Science

Sep 2024 – Present

Brandeis University, Waltham, MA

Advisor: Subhadeep Sarkar

Bachelor of Arts, Computer Science

Aug 2020 – May 2024

Cornell University, Ithaca, NY

Washington University, St. Louis, MO

Publications

- Ke Li, Ruidong Zhang, **Boao Chen**, Siyuan Chen, Sicheng Yin, Saif Mahmud, Francois Guimbretiere, and Cheng Zhang, "[GazeTrak: Exploring Acoustic-based Eye Tracking on a Glass Frame](#)", *International Conference on Mobile Computing and Networking*, 2024
- Ke Li, Ruidong Zhang, Siyuan Chen, **Boao Chen**, Mose Sakashita, Francois Guimbretiere, and Cheng Zhang, "[EyeEcho: Continuous Facial Expression Tracking on Glasses with Minimally-obtrusive Acoustic Sensing](#)", *ACM Computer Human Interaction conference*, 2024
- Ke Li, Devansh Agarwal, Ruidong Zhang, Vipin Gunda, Tianjun Mo, Saif Mahmud, **Boao Chen**, Francois Guimbretiere, and Cheng Zhang, "[SonicID: User Identification on Smart Glasses with Acoustic Sensing](#)", *The Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 2024

Poster

- Alexander Ott, **Boao Chen**, Arpita Saha, and Subhadeep Sarkar, "[Self-Designing LSM Memory Buffer](#)", 2025 North East Database Day.

Projects

- **Self-Designing LSM Memory Buffer (Ongoing):**
Designing the optimal memory buffer implementation is critical to achieving optimal performance in any log structured merge (LSM) tree. State-of-the-art LSM engines suboptimally adopt fixed buffer configurations for unknown and dynamic workloads. This traditional design choice often leads to suboptimal performance, especially when the workload shifts over time. We propose building a self-designing LSM memory buffer that can transition between different memory buffer implementations on the fly based on changing workload characteristics. Its machine-learning-based hyperparameter optimization framework learns from historical experimental data and iteratively suggests the optimal configuration of memory buffers when workload changes are detected. Finally, the current memory buffer implementation gradually transitions toward the suggested optimal memory buffer design lazily while maintaining robust performance.
- **LSM Buffer Selection (Ongoing):**
LSM-trees are known for their ingestion-optimized feature while maintaining competitive read performance. Specifically, one of the key design decisions of LSM-trees is to keep an in-memory buffer to store and append incoming entries. The buffer is sealed, sorted, and flushed to secondary storage when full. We observe that the LSM system's ingestion speed is bounded by this memory buffer design, since the performance of any LSM system is highly dependent on the buffer's configuration, given that all incoming entries must first batch write through the buffer before flushing to secondary storage. In this project, we benchmark multiple buffer implementations under diverse workloads that vary in operation type, frequency, interleaving, and distribution. We examine each design's trade-offs in write latency, point lookup latency, range lookup latency, and memory footprint across various hardware (SSD vs. HDD) and write-ahead logging (WAL) settings. Our goal is to construct a comprehensive handbook that provides guidance in selecting the appropriate in-memory buffer configuration for specific performance targets and workload patterns.

Teaching Assistantship

- **Spring 2025:** *Database Management Systems* [COSI 127B](#), Brandeis University.
- **Fall 2024:** *Algorithms* [COSI 180A](#), Brandeis University.
- **Spring 2024:** *Applied High-Performance and Parallel Computing* [CS 5520](#), Cornell University.
- **Fall 2023:** *Foundations of Artificial Intelligence* [CS 4700](#), Cornell University.
- **Spring 2023:** *Operating Systems* [CS 4410](#), Cornell University.

Certification

- **June 2025:** *Oracle Certified Professional, MySQL 8.0 Database Administrator* — [\[link\]](#)
- **June 2025:** *Oracle Certified Associate, MySQL Implementation* — [\[link\]](#)

Technical Skills

- **Programming Languages:** Python, C/C++, Java, SQL, R, MATLAB
- **Databases:** RocksDB, MariaDB, Postgres
- **Tools:** PyTorch, TensorFlow, OpenCV

Curricular Activities

- **Oct 2023:** Participated in the [Cornell Trading Competition](#) as part of a team of four students and won first place in the Algorithmic Derivative Trading Case out of 47 teams.

Society Member

Member, Association for Computing Machinery (ACM)